

A Four-variant feature decomposition

variant	$\mathbf{x}[3:6]$	unit direction $\mathbf{x}[8]$	scalar magnitude	interpretation
withleak	$\hat{\mathbf{u}}_{\text{CFD}}$ (target)		$\ \mathbf{u}_{\text{CFD}}\ /u_c$ (target)	both channels leaked
leak_dir_only	$\hat{\mathbf{u}}_{\text{CFD}}$ (target)	local radius / L_c (geom.)		direction leak only → matches withleak
leak_mag_only	PCA tangent (geom.)		$\ \mathbf{u}_{\text{CFD}}\ /u_c$ (target)	magnitude leak only → worse than noleak
noleak	PCA tangent (geom.)	local radius / L_c (geom.)		honest geometry-only baseline

target leakage

geometric proxy

B 46-aorta VMR cohort (split seed 22, frozen 2026-04-30)

46 patient-specific aortic CFD cases (25 MR + 21 CT)



5 test cases:

